

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Project Report

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Tools Used: Python | PostgreSQL | Power BI

Dataset: 3,900 Customer Transactions | 18 Features

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1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

The analysis was conducted for a leading retail company that observed shifts in purchasing patterns across demographics, product categories, and sales channels. The management team sought to understand which factors — discounts, reviews, seasons, or payment preferences — drive consumer decisions and repeat purchases.

Overarching Business Question

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

2. Dataset Summary

Attribute	Details
Total Rows	3,900
Total Columns	18
Missing Values	37 in Review Rating column
Age Range	18 – 70 years
Purchase Amount Range	\$20 – \$100 USD
Review Rating Range	2.5 – 5.0
Categories	Clothing, Footwear, Accessories, Outerwear

Key Feature Groups

Feature Group	Columns Included
Customer Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping Behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency, Review Rating, Shipping Type
Payment	Payment Method

3. Exploratory Data Analysis (Python)

Data preparation and cleaning was performed in Python using the pandas library. The following steps were carried out:

3.1 Data Cleaning & Preparation

Data Loading & Initial Exploration

- Imported the dataset using pandas
- Used `df.info()` to inspect data types and `df.describe()` for summary statistics
- Identified 37 missing values in the Review Rating column

Missing Data Handling

- Imputed missing Review Rating values using the median rating of each product category
- This approach preserves category-specific rating distributions

Column Standardization

- Renamed all columns to snake_case for consistency and readability

3.2 Feature Engineering

Age Group Column

Customer ages were binned into four quartile-based segments using `pd.qcut()`:

Age Group	Description
Young Adult	Lowest age quartile
Adult	Second quartile
Middle-Aged	Third quartile
Senior	Highest age quartile

Purchase Frequency Days

- Created a `purchase_frequency_days` column derived from the Frequency of Purchases field

Data Consistency Check

- Investigated whether `discount_applied` and `promo_code_used` were redundant

- Confirmed redundancy and dropped promo_code_used to reduce noise

3.3 Database Integration

After cleaning and feature engineering, the Python script was connected to PostgreSQL using SQLAlchemy, and the cleaned DataFrame was loaded into the database as the customer table for downstream SQL analysis.

4. Data Analysis using SQL

Structured analysis was performed in PostgreSQL to answer 10 key business questions. Results are presented below with query summaries and findings.

Q1. Revenue by Gender

Gender	Total Revenue (USD)
Female	\$75,191
Male	\$157,890

- Male customers generated approximately 2x the revenue of female customers, largely due to higher customer count

Q2. High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount (~\$59.76), indicating that discounts do not significantly reduce per-transaction value for these buyers.

Q3. Top 5 Products by Average Review Rating

Rank	Product	Avg. Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

- Ratings are tightly clustered between 3.78 – 3.86, suggesting consistently moderate customer satisfaction

Q4. Shipping Type Comparison

Shipping Type	Avg. Purchase Amount (USD)
Standard	\$58.46
Express	\$60.48

- Express shipping users spend slightly more on average, suggesting higher-value purchases tend to come with urgency

Q5. Subscribers vs. Non-Subscribers

Subscription Status	Total Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

- Average spend is nearly identical across both groups (~\$59.49 vs \$59.87)
- Non-subscribers generate more total revenue simply due to higher customer count (73% of base)
- Recommendation: Focus on converting non-subscribers — the revenue potential is significant

Q6. Most Discount-Dependent Products

Rank	Product	Discount Rate (%)
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

- These products may struggle to sell at full price and carry high dependency on promotional pricing

Q7. Customer Segmentation

Segment	Definition	Count
New	1 previous purchase	83
Returning	2–10 previous purchases	701
Loyal	11+ previous purchases	3,116

- 80% of customers are Loyal, indicating strong retention but also a need to grow the new customer funnel

Q8. Top 3 Products per Category

Category	#1 Product	#2 Product	#3 Product
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)

Category	#1 Product	#2 Product	#3 Product
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	Hoodie (154)

Q9. Repeat Buyers & Subscriptions

Subscription Status	Repeat Buyers (>5 Purchases)
No	2,518
Yes	958

- Even among repeat buyers, the majority (72%) are not subscribed — a strong opportunity to convert loyal shoppers into subscribers

Q10. Revenue by Age Group

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-Aged	\$59,197
Adult	\$55,978
Senior	\$55,763

- Young Adults contribute the most revenue, making them a high-priority target demographic

5. Power BI Dashboard

An interactive dashboard was built in Power BI to present insights visually for stakeholder decision-making. The dashboard includes dynamic filters and charts covering key dimensions of the analysis.

Dashboard KPIs

KPI	Value
Total Customers	3,900
Average Purchase Amount	\$59.76
Average Review Rating	3.75
Subscription Rate	27%

Dashboard Features

- Subscription status filter (Yes / No)
- Gender filter (Male / Female)
- Category filter (Accessories, Clothing, Footwear, Outerwear)
- Shipping type filter
- Revenue by Category — bar chart
- Sales by Category — bar chart
- Revenue by Age Group — horizontal bar chart
- Sales by Age Group — horizontal bar chart
- % Customers by Subscription Status — donut chart (27% Yes, 73% No)

6. Business Recommendations

#	Recommendation	Rationale
1	Boost Subscriptions	73% of the customer base is unsubscribed. Since average spend is equal, converting even 10% would add ~\$17,000 in tracked revenue.
2	Customer Loyalty Programs	3,116 customers are Loyal; reward programs can deepen engagement and increase basket size.
3	Review Discount Policy	Products like Hat and Sneakers have ~50% discount rates. Reassess if margin justifies the promotion frequency.
4	Product Positioning	Highlight Gloves, Sandals, and Boots in marketing — they carry the highest review ratings.
5	Targeted Marketing for Young Adults	Young Adults contribute the highest revenue (\$62,143). Tailor campaigns to this segment.
6	Convert Repeat Buyers to Subscribers	72% of repeat buyers (>5 purchases) are unsubscribed. A loyalty-linked subscription incentive could convert them.

7. Conclusion

This end-to-end analysis of customer shopping behavior reveals several actionable patterns:

- The customer base is dominated by Loyal, repeat buyers — the company has strong retention but must invest in acquiring new customers.
- Subscription rates remain low despite equal average spend, presenting a clear growth lever.
- Young Adults are the highest-revenue age group and should be prioritized in digital marketing strategies.
- Several products are heavily discount-dependent, suggesting a need to evaluate promotional strategies and product positioning.
- Top-rated products (Gloves, Sandals, Boots) do not overlap with top discount-dependent products — reinforcing that quality products can stand on their own without aggressive discounting.

Together, the Python EDA, SQL analysis, and Power BI dashboard provide a comprehensive, data-driven foundation for the retail company to optimize engagement, revenue, and customer satisfaction.